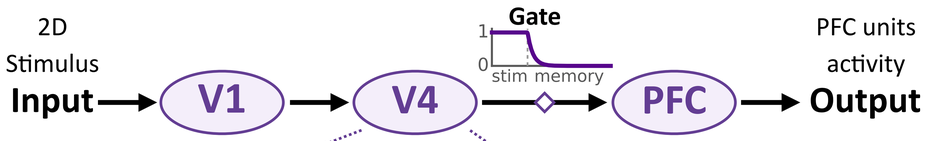
A neural circuit theory with recurrent normalization explains the dynamics of visual working memory

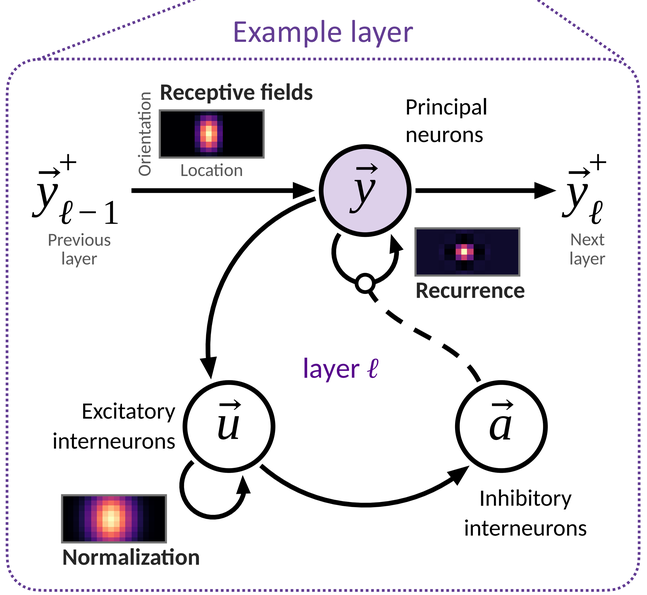
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[1]Dept of Psychology; [2]Centre for Neural Science; [3]Dept of Physics, New York University

Motivation: Visual working memory (WM) precision declines with delay, set size, and shorter stimulus duration, but no biologically grounded circuit has explained all of them together.

Conclusion: We developed a neural circuit theory that implements divisive normalization through its recurrence to fit psychophysical measurements of visual working memory dynamics.

ORGaNICs: stochastic neural circuit model[5, 6]





Membrane potential

$$\tau_y \dot{\mathbf{y}} = -\mathbf{y} + b_y \mathbf{z} + \frac{1}{1+\mathbf{a}^+} \odot \mathbf{W}\sqrt{\mathbf{y}^+} + \boldsymbol{\eta}_y$$

$$\tau_u \dot{\mathbf{u}} = -\mathbf{u} + (b_u \sigma)^2 + \mathbf{N}\left[\mathbf{y}^+ \odot (\mathbf{u}^+)^2\right] + \boldsymbol{\eta}_u$$

$$\tau_a \dot{\mathbf{a}} = -\mathbf{a} + \mathbf{u}^+ + \mathbf{a}^+ \odot \mathbf{u}^+ + \alpha \dot{\mathbf{u}} + \boldsymbol{\eta}_a$$

Firing rate

$$\tau_y \dot{\mathbf{y}}^+ = -\mathbf{y}^+ + \lfloor \mathbf{y} \rfloor^2$$

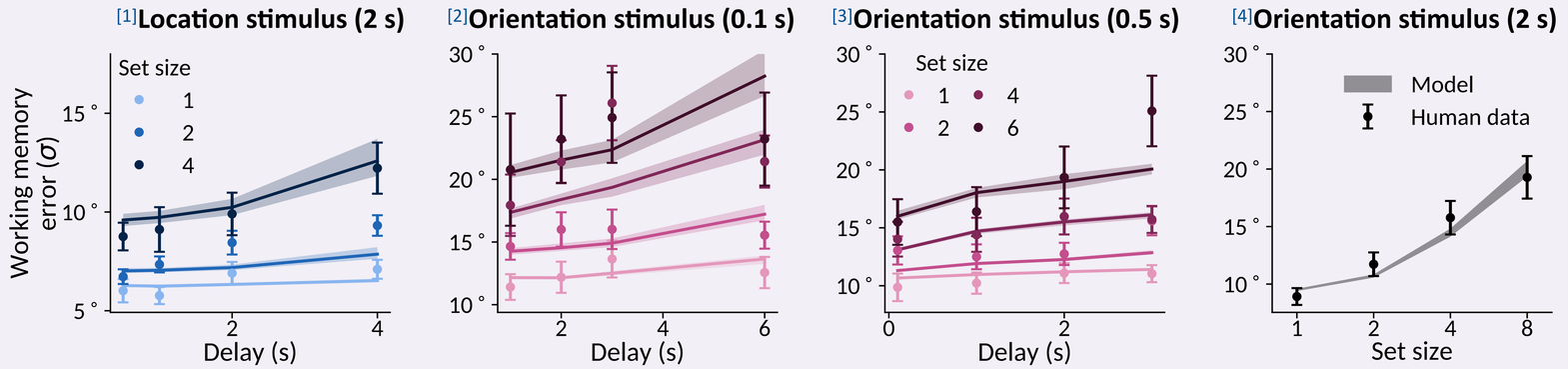
$$\tau_u \dot{\mathbf{u}}^+ = -\mathbf{u}^+ + \sqrt{\lfloor \mathbf{u} \rfloor}$$

$$\tau_a \dot{\mathbf{a}}^+ = -\mathbf{a}^+ + \lfloor \mathbf{a} \rfloor$$

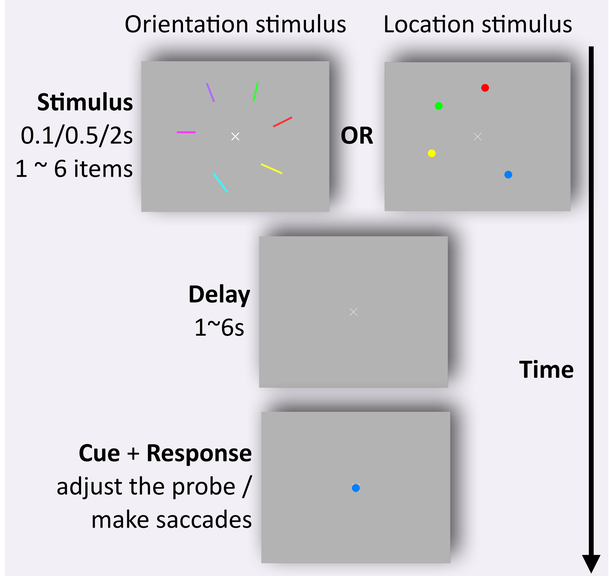
Low-pass filtered noise

$$\tau_\eta \dot{\boldsymbol{\eta}} = -\boldsymbol{\eta} + \rho\sqrt{\tau_\eta}\,\mathcal{N}(0, dt)$$

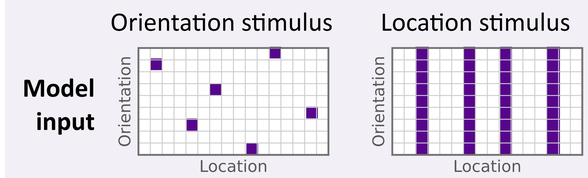
Model fits experimental measurements of spatial and featural WM performance



Working memory task

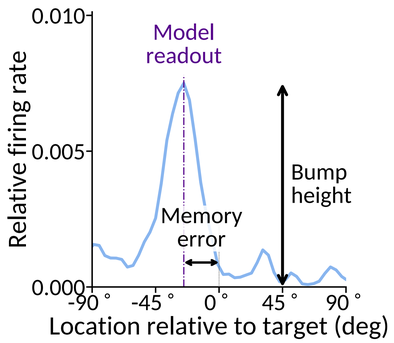


Convert stimulus to 2D model input

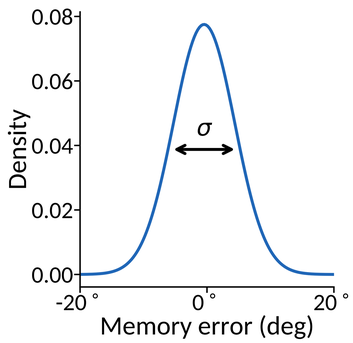


Circuit components explain visual WM dynamics

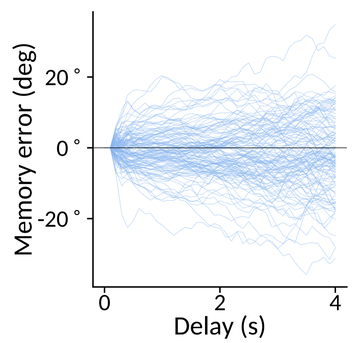
Ideal observer readout from PFC activity



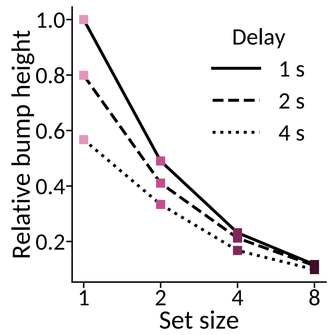
Memory error distribution



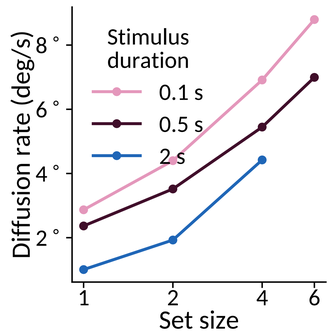
Memory errors diffuse over time[7]



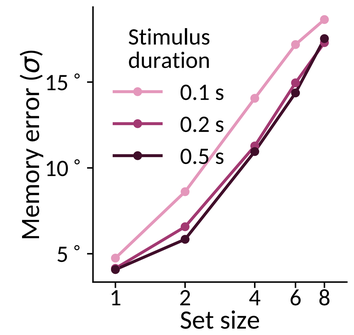
Divisive normalization weakens each bump with more items



Weaker bumps drift faster



Shorter stimulus duration, larger memory errors



Ref. [1] Schneegans & Bays, *J. Neurosci.*, (2018); [2] Shin et al., *J. Vis.*, (2017); [3] Pertzov et al., J. *Exp. Psychol. Learn. Mem. Cogn.*, (2017); [4] Bays, *J. Neurosci.*, (2014); [5] Heeger & Mackey, *PNAS*, (2019); [6] Pal et al., *eLife*, (2026); [7] Einstein, *Ann. Phys.*, (1905).